Applied Anatomy, Physiology and Physics MCQ – Chapter 1

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 1: Fundamental Concepts (Chapter 1)

50 Questions • 35 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. The larynx extends from the root of the tongue to the trachea and lies opposite which vertebral levels? **A.** C1 to C3 **B.** C3 to C6 **C.** C5 to T1 **D.** C6 to T2	**Ans: B**
Q2. What is the distance between the teeth and the vocal cords? **A.** 12–15 cm **B.** 10–15 cm **C.** 16–20 cm **D.** 5–8 cm	**Ans: A**
Q3. Which of the following is an UNPAIRED cartilage of the larynx? **A.** Arytenoid **B.** Corniculate **C.** Cuneiform **D.** Cricoid	**Ans: D**
Q4. Which is the narrowest part of the larynx in adults? **A.** The glottis **B.** The subglottis **C.** The cricoid ring **D.** The epiglottis	**Ans: A**
Q5. All muscles of the larynx are supplied by the recurrent laryngeal nerve EXCEPT which muscle? **A.** Posterior cricoarytenoid **B.** Lateral cricoarytenoid **C.** Cricothyroid **D.** Thyroarytenoid	**Ans: C**
Q6. Sensory supply above the vocal cords is provided by which nerve? **A.** Recurrent laryngeal nerve **B.** Internal branch of superior laryngeal nerve **C.** External branch of superior laryngeal nerve **D.** Glossopharyngeal nerve	**Ans: B**
Q7. In Semon's law, in bilateral partial paralysis of the recurrent laryngeal nerve, which muscles are affected first? **A.** The adductors **B.** The cricothyroid **C.** The abductors (posterior cricoarytenoids) **D.** The tensors	**Ans: C**

Q8. During general anesthesia with muscle relaxants, the vocal cords are held in which position? **A.** Fully adducted position **B.** Fully abducted position **C.** Cadaveric (mid) position **D.** Paramedian position	**Ans: C**

Q9. What is the length of the trachea? **A.** 5–8 cm **B.** 10–12 cm **C.** 15–18 cm **D.** 20–22 cm	**Ans: B**

Q10. The trachea consists of how many incomplete rings? **A.** 10–12 **B.** 12–14 **C.** 16–20 **D.** 22–24	**Ans: C**

Q11. What is the distance of the carina from the upper incisors? **A.** 28–30 cm **B.** 20–22 cm **C.** 15–18 cm **D.** 32–35 cm	**Ans: A**

Q12. Up to what age is the angle of both right and left bronchus the same (55°) in children? **A.** 1 year **B.** 2 years **C.** 3 years **D.** 5 years	**Ans: C**

Q13. What is the total number of alveoli and their surface area? **A.** 100 million, 40 m^2 **B.** 300 million, 70 m^2 **C.** 500 million, 100 m^2 **D.** 200 million, 50 m^2	**Ans: B**

Q14. Which cells of the alveolar epithelium secrete surfactant? **A.** Type I cells **B.** Type II cells **C.** Clara cells **D.** Goblet cells	**Ans: B**

Q15. According to Table 1.1, what is the length and angle of the right main bronchus? **A.** Length 5 cm, angle 45° **B.** Length 2.5 cm, angle 25° **C.** Length 2.5 cm, angle 45° **D.** Length 5 cm, angle 25°	**Ans: B**
Q16. In the supine position, the most commonly involved segment in foreign body aspiration is: **A.** Posterior basal segment of right lower lobe **B.** Posterior segment of upper lobe **C.** Apical segment of right lower lobe **D.** Lateral basal segment of left lower lobe	**Ans: C**
Q17. Ciliary activity of the respiratory mucosa is inhibited by all inhalational agents EXCEPT: **A.** Halothane **B.** Ether **C.** Isoflurane **D.** Sevoflurane	**Ans: B**
Q18. Sympathetic supply to the bronchial tree (T2 to T5) causes: **A.** Bronchoconstriction **B.** Bronchodilatation **C.** Increased secretions **D.** No effect on airway caliber	**Ans: B**
Q19. The pneumotaxic center, which has an inhibitory effect on the apneustic center, is located in the: **A.** Upper pons **B.** Lower pons **C.** Medulla **D.** Midbrain	**Ans: A**
Q20. Central respiratory centers are most sensitive to changes in: **A.** Arterial pO_2 **B.** CSF pH (influenced by pCO_2) **C.** Blood temperature **D.** Plasma bicarbonate	**Ans: B**
Q21. Peripheral chemoreceptors present in the carotid body and aortic arch are very sensitive to: **A.** Hypercarbia **B.** Acidosis **C.** Hypoxia **D.** Hyperthermia	**Ans: C**

Q22. Which is the most important muscle of inspiration, moving 1.5 cm in quiet respiration and 6–10 cm in deep breathing? **A.** External intercostals **B.** Scalene muscles **C.** Diaphragm **D.** Sternomastoids	**Ans: C**
Q23. During anesthesia with inhalational agents, expiration becomes active and is mediated by which muscles? **A.** Abdominal muscles **B.** External intercostals **C.** Diaphragm **D.** Scalene muscles	**Ans: A**
Q24. Maximum airway resistance to airflow occurs in which part of the airway? **A.** Terminal bronchi **B.** Trachea and then main bronchus **C.** Alveolar ducts **D.** Segmental bronchi	**Ans: B**
Q25. What is the normal average ventilation/perfusion (V/Q) ratio? **A.** 0.3 **B.** 2.1 **C.** 0.8 **D.** 1.5	**Ans: C**
Q26. The normal alveolar to arterial (A-a) oxygen difference is: **A.** 3–5 mm Hg **B.** 10–15 mm Hg **C.** 0.3–0.5 mm Hg **D.** 20–25 mm Hg	**Ans: A**
Q27. What is the normal anatomical dead space (as a fraction of tidal volume)? **A.** 10% of tidal volume **B.** 30% of tidal volume **C.** 50% of tidal volume **D.** 60% of tidal volume	**Ans: B**
Q28. Anatomical dead space is DECREASED by which of the following? **A.** Neck extension **B.** Atropine **C.** Intubation and tracheostomy **D.** Anesthesia mask and circuits	**Ans: C**

Q29. What is the volume of alveolar dead space in the standing position? **A.** Zero **B.** 60–80 mL **C.** 150 mL **D.** 2 mL/kg	**Ans: B**

Q30. Endotracheal tubes and tracheostomy affect anatomical dead space by: **A.** Increasing it by adding circuit volume **B.** Decreasing it by bypassing the upper airways **C.** Having no effect on it **D.** Doubling it	**Ans: B**

Q31. What is the normal oxygen uptake? **A.** 1000 mL/min **B.** 250 mL/min **C.** 500 mL/min **D.** 100 mL/min	**Ans: B**

Q32. What volume of oxygen does 1 gram of hemoglobin carry? **A.** 0.003 mL **B.** 1.34 mL **C.** 20 mL **D.** 0.34 mL	**Ans: B**

Q33. At a pO2 of 60 mm Hg, the oxygen saturation of hemoglobin is approximately: **A.** 97% **B.** 90% **C.** 75% **D.** 50%	**Ans: B**

Q34. What is the P50 (partial pressure at which oxygen saturation is 50%) and is it affected by anesthetics? **A.** 26 mm Hg; not affected by anesthetics **B.** 40 mm Hg; affected by anesthetics **C.** 60 mm Hg; not affected by anesthetics **D.** 50 mm Hg; affected by anesthetics	**Ans: A**

Q35. According to the Bohr effect, alkalosis shifts the oxygen dissociation curve: **A.** To the right **B.** To the left **C.** Upward **D.** Downward	**Ans: B**

Q36. Which of the following shifts the oxygen dissociation curve to the RIGHT? **A.** Carbon monoxide poisoning **B.** Hypothermia **C.** Inhalational anesthetics **D.** Decreased 2,3 DPG	**Ans: C**

Q37. What is the oxygen flux (amount of oxygen leaving the left ventricle per minute)? **A.** 250 mL/min **B.** 1000 mL/min **C.** 500 mL/min **D.** 5000 mL/min	**Ans: B**

Q38. In what form is the major portion of carbon dioxide transported in blood? **A.** As bicarbonate (90%) **B.** Dissolved (5%) **C.** As carbamino compounds **D.** Attached to hemoglobin as carboxyhemoglobin	**Ans: A**

Q39. Tracheal tug, a downward movement of the trachea during deep inspiration, is seen in: **A.** Flail chest **B.** Deep anesthesia with inhalational agents and partially curarized patients **C.** Hiccups **D.** Bronchospasm	**Ans: B**

Q40. Hypoxic pulmonary vasoconstriction (HPV) is maximally blunted by which inhalational agent? **A.** Ether **B.** Nitrous oxide **C.** Halothane **D.** Sevoflurane	**Ans: C**

Q41. What is the normal tidal volume? **A.** 400–500 mL (10 mL/kg) **B.** 1200–1500 mL **C.** 2400–2600 mL **D.** 100–150 mL	**Ans: A**

Q42. Functional residual capacity (FRC) is the sum of which two volumes? **A.** Inspiratory reserve volume + tidal volume **B.** Tidal volume + expiratory reserve volume **C.** Expiratory reserve volume + residual volume **D.** Inspiratory reserve volume + residual volume	**Ans: C**

Q43. During general anesthesia, the functional residual capacity (FRC) decreases by approximately: **A.** 5% **B.** 15–20% **C.** 30–40% **D.** 50%	**Ans: B**
Q44. Which lung volumes/capacities CANNOT be measured by spirometry? **A.** Tidal volume and vital capacity **B.** Inspiratory and expiratory reserve volume **C.** Residual volume, functional residual capacity and total lung capacity **D.** Inspiratory capacity	**Ans: C**
Q45. A normal breath-holding time on the simple bedside test is: **A.** > 25 seconds **B.** 15–25 seconds (borderline) **C.** < 15 seconds **D.** 5–10 seconds	**Ans: A**
Q46. Closing capacity (the volume at which the airway closes) is normally: **A.** 1 litre more than functional residual capacity **B.** 1 litre less than functional residual capacity **C.** Equal to functional residual capacity **D.** Equal to total lung capacity	**Ans: B**
Q47. What is the normal total lung compliance (lung and chest wall acting in opposite directions)? **A.** 0.2 L/cm H_2O **B.** 0.1 L/cm H_2O **C.** 0.4 L/cm H_2O **D.** 2.5 L/cm H_2O	**Ans: B**
Q48. According to Boyle's law, at a constant temperature the volume of a gas is: **A.** Directly proportional to pressure **B.** Inversely proportional to pressure **C.** Directly proportional to temperature **D.** Inversely proportional to temperature	**Ans: B**
Q49. Laminar flow is more dependent on which property, according to the Hagen-Poiseuille law? **A.** Density **B.** Temperature **C.** Viscosity **D.** Molecular weight	**Ans: C**

Q50. The Poynting effect explains which phenomenon used in the Entonox cylinder? **A.** Mixing of liquid nitrous oxide at high pressure with oxygen leads to formation of gaseous nitrous oxide **B.** Oxygen liquefies when mixed with nitrous oxide **C.** Nitrous oxide separates from oxygen at high pressure **D.** Both gases remain in liquid state in the cylinder	**Ans: A**

Answer Key & Explanations

Q1. Answer: B — C3 to C6

Page 3: The larynx is the organ of voice extending from the root of the tongue to the trachea and lies opposite C3 to C6.

Topic: Airway Anatomy

Q2. Answer: A — 12–15 cm

Page 3: The distance between teeth and vocal cords is 12–15 cm, and the distance between vocal cords and carina is 10–15 cm.

Topic: Airway Anatomy

Q3. Answer: D — Cricoid

Page 3: The larynx consists of 3 paired cartilages (arytenoid, corniculate, cuneiform) and 3 unpaired cartilages (thyroid, cricoid and epiglottis).

Topic: Airway Anatomy

Q4. Answer: A — The glottis

Page 3: The glottis is the narrowest part in adults, while the subglottis (at the level of cricoid) is the narrowest part in children up to the age of 6 years.

Topic: Airway Anatomy

Q5. Answer: C — Cricothyroid

Page 3: All muscles of the larynx are supplied by the recurrent laryngeal nerve except cricothyroid, which is supplied by the external branch of the superior laryngeal nerve.

Topic: Larynx Nerve Supply

Q6. Answer: B — Internal branch of superior laryngeal nerve

Page 3: Sensory supply up to the vocal cords is by the internal branch of the superior laryngeal nerve, and below the vocal cords by the recurrent laryngeal nerve.

Topic: Larynx Nerve Supply

Q7. Answer: C — The abductors (posterior cricoarytenoids)

Page 4: In bilateral partial paralysis of recurrent laryngeal nerve, the abductors (posterior cricoarytenoids) are affected first (Semon's Law), causing the cords to be held in adduction producing respiratory stridor.

Topic: Laryngeal Nerve Paralysis

Q8. Answer: C — Cadaveric (mid) position

Page 4: In complete paralysis of both recurrent and superior laryngeal nerves, the cords are held in mid-position (cadaveric position). The cords are also in cadaveric position during anesthesia with muscle relaxants.

Topic: Laryngeal Nerve Paralysis

Q9. Answer: B — 10–12 cm

Page 4: The length of the trachea is 10–12 cm. It starts from the cricoid ring (C6) and ends at the carina (T5).

Topic: Trachea

Q10. Answer: C — 16–20

Page 4: The trachea consists of 16–20 incomplete rings. The diameter of the trachea is 1.2 cm.

Topic: Trachea

Q11. Answer: A — 28–30 cm

Page 4: At the carina the trachea divides into right and left main bronchus. The distance of the carina from the upper incisors is 28–30 cm.

Topic: Bronchial Tree

Q12. Answer: C — 3 years

Page 4: In children the angle of both right and left bronchus is the same, i.e. 55° up to the age of 3 years. After this the right bronchus becomes shorter, wider and less acute.

Topic: Bronchial Tree

Q13. Answer: B — 300 million, 70 m²

Page 4: The total number of alveoli is 300 million with a surface area of 70 m². The thickness of the alveolar capillary membrane is 0.3 mm.

Topic: Bronchial Tree

Q14. Answer: B — Type II cells

Page 4: The alveolar epithelium has type I and type II cells. Type II cells secrete surfactant.

Topic: Bronchial Tree

Q15. Answer: B — Length 2.5 cm, angle 25°

Page 5, Table 1.1: The right bronchus is shorter (length 2.5 cm), wider, and the angle with vertical is 25°. The left bronchus is longer (5 cm), narrower, with an angle of 45°.

Topic: Bronchial Tree

Q16. Answer: C — Apical segment of right lower lobe

Page 4: Due to the shorter, wider and less acute angle of the right bronchus, foreign body aspiration is more common on the right side. In supine position the most commonly involved segment is the apical segment of right lower lobe.

Topic: Foreign Body Aspiration

Q17. Answer: B — Ether

Page 5: The mucosa has ciliated epithelium up to terminal bronchioles. Ciliary activity is inhibited by all inhalational agents except ether.

Topic: Respiratory Mucosa

Q18. Answer: B — Bronchodilatation

Page 6: Nerve supply is parasympathetic by vagus (causes bronchoconstriction) and sympathetic by T2 to T5 (causes bronchodilatation).

Topic: Bronchial Nerve Supply

Q19. Answer: A — Upper pons

Page 6: The pneumotaxic center is in the upper pons and the apneustic center is in the lower pons. The pneumotaxic center normally has an inhibitory effect on the apneustic center.

Topic: Regulation of Respiration

Q20. Answer: B — CSF pH (influenced by pCO_2)

Page 6: Central respiratory centers are most sensitive to changes in CSF pH, which is in turn influenced by pCO_2. Increase in pCO_2 stimulates respiration while decrease in pCO_2 inhibits respiration.

Topic: Regulation of Respiration

Q21. Answer: C — Hypoxia

Page 6: Peripheral chemoreceptors are present in the carotid body and aortic arch. Carotid body receptors are very sensitive to hypoxia.

Topic: Regulation of Respiration

Q22. Answer: C — Diaphragm

Page 6: The diaphragm is the most important muscle of inspiration (moves 1.5 cm in quiet respiration and 6–10 cm in deep breathing).

Topic: Muscles of Respiration

Q23. Answer: A — Abdominal muscles

Page 6: Expiration is normally passive. During anesthesia with inhalational agents, expiration is active, mediated by abdominal muscles.

Topic: Muscles of Respiration

Q24. Answer: B — Trachea and then main bronchus

Page 6: Maximum airway resistance to airflow occurs in trachea and then main bronchus, and minimum in terminal bronchi. Turbulent flow (seen in trachea and main bronchi) resistance is proportional to the square of flow rates.

Topic: Airway Resistance

Q25. Answer: C — 0.8

Page 6: Both ventilation and perfusion are more at bases compared to apex, but perfusion at base is comparatively higher, decreasing V/Q ratio towards the base (from 2.1 at apex to 0.3 at base, average 0.8).

Topic: Ventilation/Perfusion

Q26. Answer: A — 3–5 mm Hg

Page 6: The V/Q mismatch creates alveolar to arterial [(A-a) pO_2 difference] which is normally 3–5 mm Hg. This A-a difference is increased in lung pathologies affecting alveoli.

Topic: Ventilation/Perfusion

Q27. Answer: B — 30% of tidal volume

Page 7: Anatomical dead space is constituted by air present in nose, trachea and bronchial tree (up to terminal bronchioles). Normally it is 30% of tidal volume or 2 mL/kg or 150 mL.

Topic: Dead Space

Q28. Answer: C — Intubation and tracheostomy

Page 7: Anatomical dead space is decreased by intubation (nasal cavity bypassed if tube diameter is less than airway diameter), tracheostomy (upper airways and nasal cavity bypassed), hyperventilation, neck flexion and bronchoconstrictors. It is increased by old age, neck extension, jaw protrusion, bronchodilators, atropine, and anesthesia mask/circuits.

Topic: Dead Space

Q29. Answer: B — 60–80 mL

Page 7: Alveolar dead space is constituted by alveoli which are only ventilated but not perfused. It is 60–80 mL in standing position and zero in lying position (perfusion is equal in all parts of lung when lying).

Topic: Dead Space

Q30. Answer: B — Decreasing it by bypassing the upper airways

Page 7: Endotracheal tubes and tracheostomy decrease the anatomical dead space by bypassing the upper airways. However, all anesthesia circuits, masks and humidifiers increase the anatomical dead space.

Topic: Anesthesia and Dead Space

Q31. Answer: B — 250 mL/min

Page 7: Normal oxygen uptake is 250 mL/min. Oxygen is mainly carried in blood attached to hemoglobin (1 g of Hb carries 1.34 mL of oxygen).

Topic: Oxygen in Blood

Q32. Answer: B — 1.34 mL

Page 7: 1 g of Hb carries 1.34 mL of oxygen. A very small amount, 0.003 mL/dL/mm Hg, is carried as dissolved fraction. Oxygen content of arterial blood is 20 mL/dL and that of venous blood is 15 mL/dL.

Topic: Oxygen in Blood

Q33. Answer: B — 90%

Page 7: Normally Hb is 97% saturated at normal pO_2 of 95–98 mm Hg. At 60 mm Hg saturation is still 90%. After this point there is a sudden drop in oxygen saturation (cyanosis appears when pO_2 falls below 50–60 mm Hg).

Topic: Oxygen Dissociation Curve

Q34. Answer: A — 26 mm Hg; not affected by anesthetics

Page 8: P50 is the partial pressure at which oxygen saturation is 50%. The partial pressure of oxygen for 50% saturation is 26 mm Hg. P50 is not affected by anesthetics.

Topic: Oxygen Dissociation Curve

Q35. Answer: B — To the left

Page 8: Bohr effect — alkalosis shifts the O_2 dissociation curve to the left and acidosis to the right.

Topic: Oxygen Dissociation Curve

Q36. Answer: C — Inhalational anesthetics

Page 8: Shift to the right is seen with acidosis, high pCO_2, increased 2,3 DPG, hyperthermia and inhalational anesthetics. Shift to the left is seen with alkalosis, low pCO_2, decreased 2,3 DPG, carbon monoxide poisoning, abnormal hemoglobins, hypophosphatemia and hypothermia.

Topic: Oxygen Dissociation Curve

Q37. Answer: B — 1000 mL/min

Page 8: Oxygen flux is the amount of oxygen leaving the left ventricle per minute. It is 1,000 mL/min.

Topic: Oxygen in Blood

Q38. Answer: A — As bicarbonate (90%)

Page 8: Carbon dioxide is transported in blood as bicarbonate (90%), dissolved (5%), and as carbamino compounds. Deoxygenated blood has more CO_2 content at a given pCO_2 — this is called the Haldane effect.

Topic: Carbon Dioxide

Q39. Answer: B — Deep anesthesia with inhalational agents and partially curarized patients

Page 8: Tracheal tug is a downward movement of trachea during deep inspiration. It is seen in deep anesthesia (by inhalational agents) and in partially curarized patients.

Topic: Abnormalities of Chest Movements

Q40. Answer: C — Halothane

Page 9: HPV is a protective mechanism — whenever there is hypoxia, vasoconstriction occurs shunting blood to well-perfused areas. All inhalational agents blunt this hypoxic pulmonary vasoconstriction (maximum with halothane).

Topic: Hypoxic Pulmonary Vasoconstriction

Q41. Answer: A — 400–500 mL (10 mL/kg)

Page 9: Tidal volume is the volume of gas inspired or expired in each breath during normal quiet respiration. It is 400–500 mL (10 mL/kg).

Topic: Lung Volumes

Q42. Answer: C — Expiratory reserve volume + residual volume

Page 9: Functional residual capacity (FRC) is the volume of gas in lungs after end expiration. It is ERV + RV and equals 2400–2600 mL. During general anesthesia FRC decreases by 15–20%.

Topic: Lung Volumes

Q43. Answer: B — 15–20%

Page 9: During general anesthesia FRC decreases by 15–20%. FRC is the volume of gas in lungs after end expiration (ERV + RV).

Topic: Lung Volumes

Q44. Answer: C — Residual volume, functional residual capacity and total lung capacity

Page 10: Spirometry cannot measure residual volume; therefore any lung volume which requires measurement of residual volume — i.e. functional residual capacity and total lung capacity — cannot be measured by spirometry. Body plethysmography, helium dilution and nitrogen washout are used for these.

Topic: Pulmonary Function Tests

Q45. Answer: A — > 25 seconds

Page 9–10: Breath-holding time is a very simple and useful bedside test. Normal is > 25 seconds. Patients with breath-holding time of 15–25 seconds are considered borderline cases and breath holding time < 15 seconds indicates severe pulmonary dysfunction.

Topic: Simple Bedside Test

Q46. Answer: B — 1 litre less than functional residual capacity

Page 10: Closing capacity is the volume at which airway closes, which is normally 1 litre less than functional residual capacity. If FRC falls below closing capacity there will be significant hypoventilation and V/Q abnormalities.

Topic: Pulmonary Function Tests

Q47. Answer: B — 0.1 L/cm H_2O

Page 10: Lung compliance is 0.2 L/cm H_2O, chest wall compliance is 0.2 L/cm H_2O, and total compliance is 0.1 L/cm H_2O (lung and chest wall compliance act in opposite direction).

Topic: Lung Compliance

Q48. Answer: B — Inversely proportional to pressure

Page 11: Boyle's law states that at a constant temperature, the volume of a gas is inversely proportional to the pressure. Charle's law states that at constant pressure, volume of gas is directly proportional to temperature.

Topic: Physics Related to Anesthesia

Q49. Answer: C — Viscosity

Page 11: Laminar flow is produced when gas passes through a straight tube and is smooth. Laminar flow is more dependent on viscosity. At laminar flow, Hagen-Poiseuille law applies (flow rate directly proportional to pressure gradient and fourth power of radius, inversely proportional to viscosity and length). Turbulent flow is more dependent on density.

Topic: Physics Related to Anesthesia

Q50. Answer: A — Mixing of liquid nitrous oxide at high pressure with oxygen leads to formation of gaseous nitrous oxide

Page 11: The Poynting effect — mixing of liquid nitrous oxide at low pressure with oxygen at high pressure (in Entonox) leads to formation of gas of nitrous oxide. Therefore oxygen and nitrous oxide are both present in gaseous state in the Entonox cylinder.

Topic: Physics Related to Anesthesia